# Usability Test Plan

## 1. Purpose

The purpose of this usability test is to evaluate the usability of the polling application, and see if users can use the poll application to complete core tasks effectively, efficiently, and satisfactorily. The test aims to identify usability problems related to different user roles when creating, joining, and participating in quizzes.

## 2. Research questions

The usability test was designed with the following questions in mind :

- Can users successfully complete the core tasks of the system?
- Do users experience usability or accessibility issues while interacting with the system?
- Do users require significant effort or time to understand how the system works?
- Can users complete the given tasks without assistance?
- Can users complete the given tasks efficiently without unnecessary steps?
- Where do users experience confusion during task completion?
- Are users satisfied with the system after completing the tasks?

## 3. Method

### 3.1 Participants characteristics

Although the application supports multiple users being online at the same time, it was run locally during testing.Therefore, simultaneous interaction had to be simulated by opening the application in multiple browser windows or tabs on the same device.

Due to these limitations, participants took part in the usability test by performing both presenter and audience roles. This allowed each participant to experience the workflow of creating a quiz as well as joining and responding to it as an audience member. Participants had general experience using web or mobile applications, but no prior experience with the application.

### 3.2 Tasks

Participants were asked to complete a set of predefined tasks representing the core functionality of the system. Tasks differed depending on the user role.

Audience tasks included:
- Switching the interface language and continuing to respond.
- Joining a quiz using a code.
- Entering name.

• Answering quiz questions.
• Identifying whether an answer was successfully submitted.
• Responding to a second question after an automatic transition.
• Viewing profile and changing application language and username.

Presenter tasks included:
• Switching the interface language and attempting to launch a quiz.
• Creating a short quiz with at least two questions.
• Launching the quiz and making it available to participants.
• Monitoring incoming responses.
• Identifying the result for each question.
• Viewing profile and changing application language and username

### 3.3 Test environment, equipment and logistics

The usability test was conducted in an informal setting using a developer's laptop, on which the application was run locally. The test was run on the developer's system. During the usability test, each participant acted as both presenter and audience to cover the full interaction flow.

The test was moderated by the developer in person. Assistance was provided whenever a participant requested for it.

### 3.4 Data to be collected

Both qualitative and quantitative data were collected through direct observation and participant feedback during the usability test.

Quantitative data consisted of whether participants were able to complete each task successfully and the number of observable errors encountered during task execution.

Qualitative data included participants' verbal comments during the test, observed moments of confusion or hesitation, and feedback collected through a short post-test interview.

### 3.5 Test moderator role

The test moderator observed participants during task completion and encouraged them to think aloud. The moderator did not provide assistance unless participants were completely blocked or unable to proceed. Assistance was also provided if it was requested.

## 4. Summary of Results

The usability test was conducted with four participants. Two participants interacted with the system using a desktop browser in a standard web view, while the other two used a mobile-sized interface simulated through the browser's developer tools.

Overall, both the web and mobile interfaces allowed participants to complete the main tasks of creating and participating in a poll without difficulties. Creating a new poll was generally smooth on both interfaces, and participants were able to understand the basic workflow without external assistance.
However, several usability issues were observed. On the mobile interface, participants initially experienced slight delays when starting to use the system, as the “Create poll” and “Participate in poll” buttons were positioned lower on the page and not fully visible without scrolling. In contrast, participants using the web interface were able to locate these options immediately.

During poll creation, one participant hesitated briefly when attempting to create a multiple-choice question, as the test setup only supported single-choice questions. Although this did not prevent task completion, it caused minor confusion and momentarily interrupted the workflow.

Additionally, during the quiz session, participants noted that even when all audience members had submitted their answers, the system did not automatically transition to the next question. Instead, the interface waited until the countdown timer ended. While this behavior was technically consistent, it was unexpected for some participants and led to short periods of uncertainty about whether further action was required.

In summary, the test results indicate that the core functionality of the system is easy to learn and use, particularly on the web interface. The identified issues primarily relate to visibility on mobile devices, clarity of supported question types during poll creation, and feedback during timed question transitions. Addressing these areas could further improve the overall user experience.